Great Salt Lake dust model

Predict PM10 dust and its arsenic moving from Great Salt Lake sediment into the Salt Lake Valley. Outputs are arrival mass and arrival rate; the model stops before health effects.

How the ODE works

For each lakebed cell **i**, **S** is PM10-emittable sediment and **I** is airborne dust. **R** accumulates dust entering the valley. All are masses in kg: sediment → airborne dust → valley arrival.

$$\beta_i(t) = K_i X_i(t) \phi_i(t) \left[\max(u_{*,i}(t) - u_{*t,i}(t), 0)\right]^p$$

$$\frac{dS_i}{dt} = G_i - \beta_i S_i \qquad \frac{dI_i}{dt} = \beta_i S_i - (k_{V,i} + k_{L,i}) I_i$$

$$\frac{dR}{dt} = \sum_i k_{V,i} I_i \qquad \frac{dR_{As}}{dt} = \sum_i c_{As,i} \eta_i k_{V,i} I_i$$

X is the exposed fraction of a cell, calculated from lake elevation and terrain. **φ** is erodibility (0–1); **u*** is friction velocity and **u*t** its erosion threshold. **G** replenishes sediment. **kV** transfers dust into the valley; **kL** removes it by deposition or transport elsewhere. β and k have inverse-time units; calibrate K accordingly. **p** controls wind sensitivity. **cAs** is kg arsenic per kg sediment; **η** adjusts for enrichment in airborne particles.

Where physics and ML fit

Prototype with the ODE. Then replace its airborne transfer equations with **HYSPLIT dispersion driven by HRRR**, using emissions βS. HYSPLIT handles transport and deposition in place of kV and kL, and provides concentrations for monitor comparisons.

Component	Real world inputs	Model choice
Exposure X	USGS Saltair lake gauge; USGS terrain / Utah lidar	Geometry; no ML
Erodibility φ	PI-SWERL field tests; moisture, crust, texture, satellite imagery	RF/XGBoost for φ with field labels; otherwise physics
Arsenic cAs and η	USGS sediment and dust chemistry; particle size matched samples	GP/kriging for cAs; measured or bounded η
Transport	NOAA HRRR winds and mixing; local weather stations	HYSPLIT dispersion; no ML
Validation	EPA AQS / Utah DAQ PM10; independent plume and filter chemistry evidence	Held out events and locations

Assumptions to test

Surface: cells use average properties; exposed ground may be crusted and non-erodible. Match elevation units and vertical datums, use lake levels by arm, and test grid resolution. Initial S and replenishment G need independent estimates.

Emission and chemistry: calibrate the approximate threshold law against field fluxes. Arsenic rides on dust with a cell-specific composition; size-dependent transport and enrichment require particle bins or uncertainty bounds.

Transport: the ODE assumes well mixed compartments and first order transfer. Count first valley entry once. R is cumulative arrival, not airborne concentration; PM10 alone cannot identify GSL dust or validate arsenic.

Build order: 1 km grid and hourly inputs → field-calibrated emissions → dispersion → held out event and chemistry validation → lake-level scenarios under identical weather. Report uncertainty ranges for emission, chemistry and transport.